CSE260 Fall 25
Assignment 1

This assignment must be hand-written. Show ALL steps in ALL questions

1. Convert the following numbers to equivalent decimal numbers.
    (a) $(100111.11011)_2$
    (b) $(3A6.B3C)_{16}$ **[Ungraded]**
    (c) $(79B.2FG)_{17}$

2. Convert the following numbers to the equivalent given base.
    (a) $(107.0425)_{10}$ to binary (Keep up to 3 decimal places)
    (b) $(6045)_7$ to base 15
    (c) $(E16)_{16}$ to base 4 **[Ungraded]**

3. Convert the following numbers to the equivalent given base. **[Ungraded]**
    (a) $(1001011011101011101111.01110010100111)_2$ to octal and hexadecimal **[Ungraded]**
    (b) $(3570.2416)_8$ to binary **[Ungraded]**
    (c) $(6CEF.7B9A)_{16}$ to binary **[Ungraded]**

4. Perform addition, subtraction and multiplication for the pair of following numbers.
    (a) $(4EB4)_{17}$ and $(9B)_{17}$
    (b) $(A81)_{12}$ and $(B98)_{12}$ **[Ungraded]**

5. Perform the following operations:

    (a) Divide $(4432)_5$ with $(13)_5$

    (b) Divide $(7763)_8$ with $(62)_8$ **[Ungraded]**

6. You are a computer enthusiast and want to buy two 8 GB DDR4 RAMs. Each RAM costs $(F1)_{16}$ dollars. You also want to buy a graphics card RTX4070Ti which costs $(10111100111)_2$ dollars. However, you don't have that much money and are afraid to ask your parents about it. Suddenly, one of your generous friends agreed to give you the money you need. He decided to give you $(3376)_8$ dollars. How much will you have left after buying those components? (Show the answer in decimal)

7. Convert the following decimal numbers to the equivalent given base.
    (a) $(472.2097)_{10}$ to Excess 6
    (b) $(10100110110110010 10.01001001)_{\text{Excess 3}}$ to Octal

8. Convert the following decimal numbers to 9 bits sign-and-magnitude, 1s complement, and 2s complement.

(a) $+(147)_{10}$

(b) $-(193)_{10}$

9. Subtract 97 from -25 using 7-bit 2's complement number system. Justify if there is an overflow. [Show calculations]